Brandon Li

978-245-5532 | jobs@brandonli.me | brandonli.me | linkedin.com/in/brandonli28 | github.com/PulseBeat02

Education

University of California, Los Angeles (UCLA) — Expected June 2028
Bachelor of Science in Computer Science — *Los Angeles, CA*

Experience

Software Development Engineer Internship (Incoming) — Sep 2026 – Dec 2026
Amazon (AWS) — *Boston, MA*

Software Engineering Intern — June 2026 – Sep 2026
Google (YouTube) — *San Bruno, CA*

- Added monochrome AV1 support to YouTube's Android app to allow thousands of more videos to be watchable.
- Added 10-bit and 12-bit HDR support to AV1 software decoding to enhance video quality and user experience.

Open Source Contributor — Jan 2026 – Present
VideoLAN — *Remote*

- Added peak limiter and expander audio filters, refactoring a shared dynamics core from the compressor.
- Integrated ggml and OpenCV dependencies into the build system (contribs) for on-device model execution.
- Ported legacy OpenCV modules to modern C++ and upgraded the SAM2 filter to SAM3 for text-prompts.
- Cut the final VLC binary size by 5% by adding linker-level dead-code stripping to eliminate unused symbols.

Software Engineering Intern — June 2025 – Sep 2025
VideoLAN — *Remote*

- Developed C/C++ video filters (real-time visual effects and processing) for VLC Media Player (traffic cone).
- Integrated a SAM2 model module with OpenCV to enable real-time object segmentation for visual analysis.
- Implemented YuNet-based ML face detection for real-time tracking of 100+ faces in surveillance systems.
- Built fast video-to-GIF conversion with 5 dithering algorithms for customizable quality and performance.
- Analyzed the AV1 video-codec specification and studied dav1d decoding algorithms to deepen knowledge.

Projects

yt-media-storage | *C++, Assembly/SIMD, FFmpeg, Coding Theory, Compression* — Feb 2026 – Present

- Popular C++ tool (890 stars, 110+ forks) that encodes local files into YouTube videos for storage.
- Achieved a 100x speed-up using inline Assembly, SIMD intrinsics, and OpenMP parallelization.
- Designed lightweight Qt 6 UI and libsodium encryption, enhancing user experience and security.
- Uses robust CRC/Wirehair error-correction to novelly bypass YouTube's compression algorithm.
- Reached #5 on Hacker News (Y Combinator) within 12 hours, generating thousands of views.
- Includes technical explanation video with over 3.3M impressions, educating over 270k viewers.

mcav | *Java, Spring Boot, TypeScript, CI/CD* — June 2020 – Present

- Popular Java framework (130+ stars, 10 forks) for building fast and portable media applications.
- Integrates with VLC/mpv with a documented image-processing pipeline API for easier adoption.
- Uses Spring Boot to stream sub-100ms latency audio to usable TypeScript/Howler.js frontend.
- Configured CI/CD pipeline with TeamCity using Oracle Cloud to automate software releases.
- Engineered native media player via JavaCV that is 5x faster than most pure-Java players.

Technical Skills

Languages: Java, C/C++, OCaml, JavaScript/TypeScript, Assembly
Frameworks & Libraries: Spring, Hibernate, FFmpeg, OpenCV, libVLC, React, Next.js, Tailwind
Technologies: Git, MongoDB, AWS, Google Cloud, Oracle Cloud
Awards: Dean's Honor List